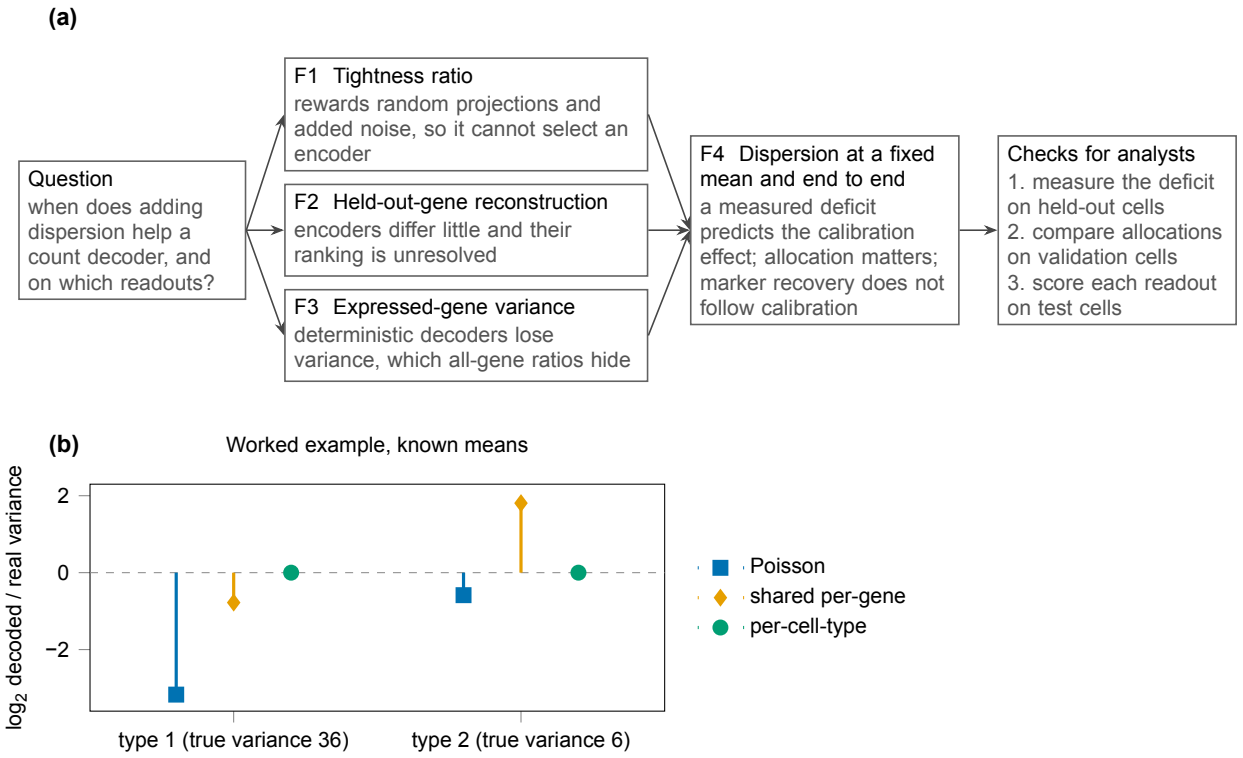
(a)
Question
when does adding dispersion help a count decoder, and on which readouts?
F1 Tightness ratio
rewards random projections and added noise, so it cannot select an encoder
F2 Held-out-gene reconstruction
encoders differ little and their ranking is unresolved
F3 Expressed-gene variance
deterministic decoders lose variance, which all-gene ratios hide
F4 Dispersion at a fixed mean and end to end
a measured deficit predicts the calibration effect; allocation matters; marker recovery does not follow calibration
Checks for analysts
1. measure the deficit on held-out cells
2. compare allocations on validation cells
3. score each readout on test cells
(b)
Worked example, known means
log2 decoded / real variance
2
0
−2
type 1 (true variance 36)
type 2 (true variance 6)
Poisson
shared per-gene
per-cell-type